

ECO387C - Fall 2024

Problem Set 2

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You may work in groups, but please acknowledge any contributions from others and turn in separate assignments.

Question 1 - Stochastic Endowment Economy

Consider the following stochastic endowment economy.

- There are two agents A and B who live for two periods ($t = 0, 1$).
- The number of each type of agent is equal ($N^A = N^B = 1$)
- At time 0, the agents receive a certain amount of endowment. However, in period 1, two possible states could be realized (state 1 and 2). The probability of each state is p and $1 - p$, respectively.
- More specifically, agents have state contingent endowments: $y^\ell = (y_0^\ell, y_1^\ell, y_2^\ell)$. For each $j \in \{A, B\}$, y_0^ℓ is the endowment received in period 0. y_1^ℓ is the endowment received in period 1, state 1, and y_2^ℓ is the endowment received in period 1, state 2.
- Facing the uncertainty about the endowment in period 1, the agents make a consumption decision at time 0. The utility functions are

$$U^\ell(c_0^\ell, c_1^\ell, c_2^\ell) = E_0 \left[\sum_{t=0}^1 (\beta^\ell)^t u^\ell(c_t^\ell) \right] = \log(c_0^\ell) + \beta^\ell \left[p \cdot \log(c_1^\ell) + (1 - p) \cdot \log(c_2^\ell) \right]$$

where β^ℓ is the discount rate for agent ℓ .

- Riskless, real bond (b^ℓ) is traded at time 0. One unit of this bond purchased in period 0 provides $(1 + r)$ units of consumption in both states in period 1. There is no bond traded in period 1 or initial bond holding.
- a Set up the agent's optimization problem. (*Hint: There are three budget constraints.*)
 - b Derive both agents' Euler equations. (*Hint: Recall the Euler equation under uncertainty discussed in class.*)
 - c Define equilibrium in this economy.

- d Solve for equilibrium price and allocation using the following parameter values. (Hint: it is not possible to avoid some tedious computations but the numbers are chosen so that you can find the equilibrium without a computer.)

$$p = \frac{1}{2}, \quad \beta^A = \frac{1}{2}, \quad \beta^B = \frac{1}{3}, \quad y^A = \left(\frac{19}{8}, 1, 3\right), \quad y^B = \left(\frac{21}{8}, 5, 3\right)$$

Question 2 - Numerical Analysis

The purpose of these questions is to further familiarize you with the numerical tools we covered in class, and to motivate computational tools we may cover in the future.

2.1 - Interpolation

- Consider the function $f(x) = 1 + x + 2x^2 - 3x^3$.
 - Generate x , a vector of 1001 linearly spaced points along $[-1, 1]$ with `linspace`.
 - Generate the vector y by applying $f(x)$ to each element of the vector above. Plot the results.
 - Create a 11-element vector of data points (`xdata`) and corresponding outputs (`ydata`).
 - Use linear and cubic interpolation to obtain the interpolated estimates of the original y (`yest`)
 - Plot the `yests` against x . Which family of basis functions was closer to the true function? Briefly explain why.
- Consider the function $f(x) = \sqrt{|x|}$.
 - Repeat the same steps as above.

The next question asks you to consider why we do not naively use the polynomial family of basis functions ($x^0, x^1, x^2 \dots$) when interpolating data. This may seem a little counter-intuitive when you consider the Stone-Weierstrass theorem.¹

- Consider the function $f(x) = \frac{1}{1+25x^2}$ over the interval $[-1, 1]$.
 - Repeat the same steps as above, except use polynomial interpolation instead (fit a 10th degree polynomial in this case) through the 11 `xdata` vector with `polyfit` and `polyeval`.
 - Plot the true function and the function generated through polynomial interpolation. What do you notice?

2.2 - Root-Finding

- Consider the demand function $q(p) = 0.1p^{-0.2} + 0.7p^{-0.5}$. What is the price that clears the market for a quantity of two units?

¹There are other reasons why polynomial interpolation on linearly spaced points is unideal in practice.

2.3 - Numerical Integration

1. The demand function for an arbitrary good is $q(p) = 2p^{-0.5}$. Numerically compute the change in consumer surplus when the price falls from 4 to 1.
2. The random variable $\tilde{X}$ is drawn from a normal distribution with mean 1.5 and variance 1.69. Using 30 nodes, numerically compute the expected value of $\sin(\tilde{X}) + \exp(\tilde{X})$.